

RD INFO TECHNOLOGY – DATA SCIENCE INTERNSHIP

TASK 2

DATA ACQUISITION

CUSTOMER CHURN PREDICTION USING MACHINE LEARNING

Submitted By

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Project Domain

Data Science & Machine Learning



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Introduction

Data acquisition is the process of collecting data from different sources for analysis and machine learning purposes. It is an important step in every data science project because the quality of collected data directly affects model performance and business insights.

In this task, customer churn data is collected from external datasets and prepared for further analysis.

Description

The objective of this task is to collect customer-related data from available sources and understand how the data can be used for machine learning and business analysis.

The collected data contains customer information such as:

- Customer demographics
- Subscription details
- Payment methods
- Internet services
- Monthly charges
- Churn status

This data will be used to analyze customer behavior and predict customer churn.

Dataset Source

The dataset used for this project is collected from:

- Kaggle
- CSV files
- Public customer churn datasets

	A	B	C	D	E	F	G	H	I	J	K	L
	CustomerID	Count	Country	State	City	Zip Code	Lat Long	Latitude	Longitude	Gender	Senior Citizen	Partn
3	9237-HQITU	1	United States	California	Los Angeles	90005	34.059281, -118.30742	34.059281	-118.307420	Female	No	No
4	9305-CDSKC	1	United States	California	Los Angeles	90006	34.048013, -118.293953	34.048013	-118.293953	Female	No	No
5	7892-POOKP	1	United States	California	Los Angeles	90010	34.062125, -118.315709	34.062125	-118.315709	Female	No	Yes
6	0280-XJGEX	1	United States	California	Los Angeles	90015	34.039224, -118.266293	34.039224	-118.266293	Male	No	No
10	6467-CHFZW	1	United States	California	Los Angeles	90028	34.099869, -118.326843	34.099869	-118.326843	Male	No	Yes
13	6047-YHPVI	1	United States	California	Los Angeles	90039	34.110845, -118.259595	34.110845	-118.259595	Male	No	No
14	5380-WKQOV	1	United States	California	Los Angeles	90041	34.137412, -118.207607	34.137412	-118.207607	Male	No	No
15	8168-UQWWF	1	United States	California	Los Angeles	90042	34.11572, -118.192754	34.115720	-118.192754	Female	No	No
16	7760-OYPDY	1	United States	California	Los Angeles	90056	33.987945, -118.370442	33.987945	-118.370442	Female	No	No
17	9420-LOJIX	1	United States	California	Los Angeles	90061	33.92128, -118.274186	33.921280	-118.274186	Female	No	No
18	7495-OOKFY	1	United States	California	Los Angeles	90063	34.044271, -118.185237	34.044271	-118.185237	Female	Yes	Yes
19	1658-BYGQY	1	United States	California	Los Angeles	90065	34.108833, -118.229715	34.108833	-118.229715	Male	Yes	No
20	5698-BQJOH	1	United States	California	Beverly Hills	90211	34.063947, -118.383001	34.063947	-118.383001	Female	No	No
21	5919-TMRGD	1	United States	California	untington Par	90255	33.97803, -118.217141	33.978030	-118.217141	Female	No	No
22	9191-MYQKX	1	United States	California	Lynwood	90262	33.923573, -118.200669	33.923573	-118.200669	Female	No	Yes
23	8637-XJIVR	1	United States	California	Marina Del Rey	90292	33.977468, -118.445475	33.977468	-118.445475	Female	No	No
25	4598-XLKNJ	1	United States	California	Inglewood	90303	33.936291, -118.332639	33.936291	-118.332639	Female	Yes	Yes
27	0486-HECZI	1	United States	California	Torrance	90504	33.867257, -118.330794	33.867257	-118.330794	Male	No	Yes
28	4846-WHAFZ	1	United States	California	Whittier	90602	33.972119, -118.020188	33.972119	-118.020188	Female	Yes	Yes
29	5299-RULOA	1	United States	California	Whittier	90606	33.976678, -118.065875	33.976678	-118.065875	Female	No	Yes
30	0404-SWRVG	1	United States	California	La Habra	90631	33.940619, -117.9513	33.940619	-117.951300	Male	No	No
31	4412-YITKF	1	United States	California	Pico Rivera	90660	33.989524, -118.089299	33.989524	-118.089299	Female	Yes	No

Tools Used

The following tools are used for data acquisition:

- Python
- Pandas
- SQL
- APIs
- MS Excel
- Kaggle
- CSV Files

Python Code

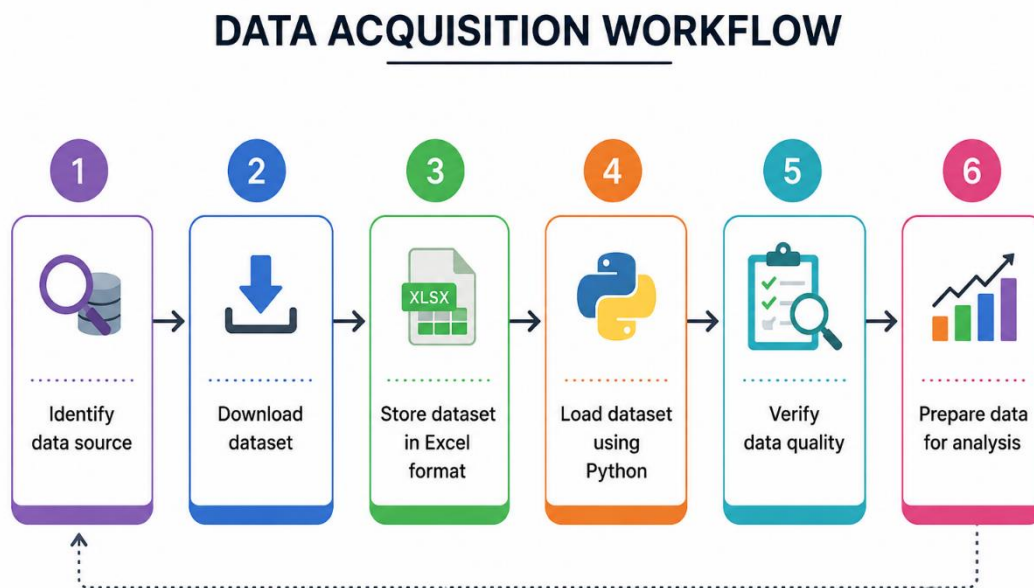
```
import pandas as pd
```

```
data = pd.read_csv('customer_churn.csv')
```

```
print(data.head())
```

Data Acquisition Workflow

1. Identify data source
2. Download dataset
3. Store dataset in excel format
4. Load dataset using Python
5. Verify data quality
6. Prepare data for analysis



Observations

- The dataset contains both numerical and categorical data.
- Customer churn information is available.
- Some missing values may require preprocessing.
- Data is suitable for machine learning classification tasks.

Conclusion

Data acquisition is an important stage in data science projects. In this task, customer churn data was successfully collected from external sources and prepared for analysis. This task helped in understanding how datasets are collected, stored, and used in machine learning projects.

Thank You

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